

Presidency

School of Artificial Intelligence and Advanced Computing



CSS7102- Mini Project Review-1

CIVIC-VOICE An AI Assistant for Multi-Channel Citizen Feedback Analysis in Public Services

Batch Number: CAI-21

Under the Supervision of,

Roll Number

20231CAI0117

20231CAI0120

20231CAI0115

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Content

- Problem Statement
- Github Link
- Analysis of Problem Statement
- Timeline of the Project
- References



Problem Statement Number: 40

Organization: Public Service Organizations (Government Agencies)

Category : Software

Problem Description:

Public agencies receive citizen feedback through emails, social media, and surveys but often analyze these in silos. An AI assistant that unifies feedback, detects themes, and highlights emerging issues would improve responsiveness. The solution should focus on sentiment and topic modeling across channels, equity-aware treatment of minority voices, and dashboards for officials to track concerns over time.

Difficulty Level: Medium–High

Github Link

Github Link: <https://github.com/inchara-k-r/AI-Assistant-for-Multi-Channel-Citizen-Feedback-Analysis>



Analysis of Problem Statement

Technology Stack Components:

Component	Technology	Purpose
Frontend	Streamlit / HTML, CSS, JavaScript	User interface and dashboard
Backend	Python, Flask (or FastAPI)	API development and application logic
Programming Language	Python	Core development
NLP & AI	Hugging Face Transformers, BERT, NLTK, spaCy	Text preprocessing, sentiment analysis, summarization
Machine Learning	Scikit-learn	Complaint classification and model evaluation

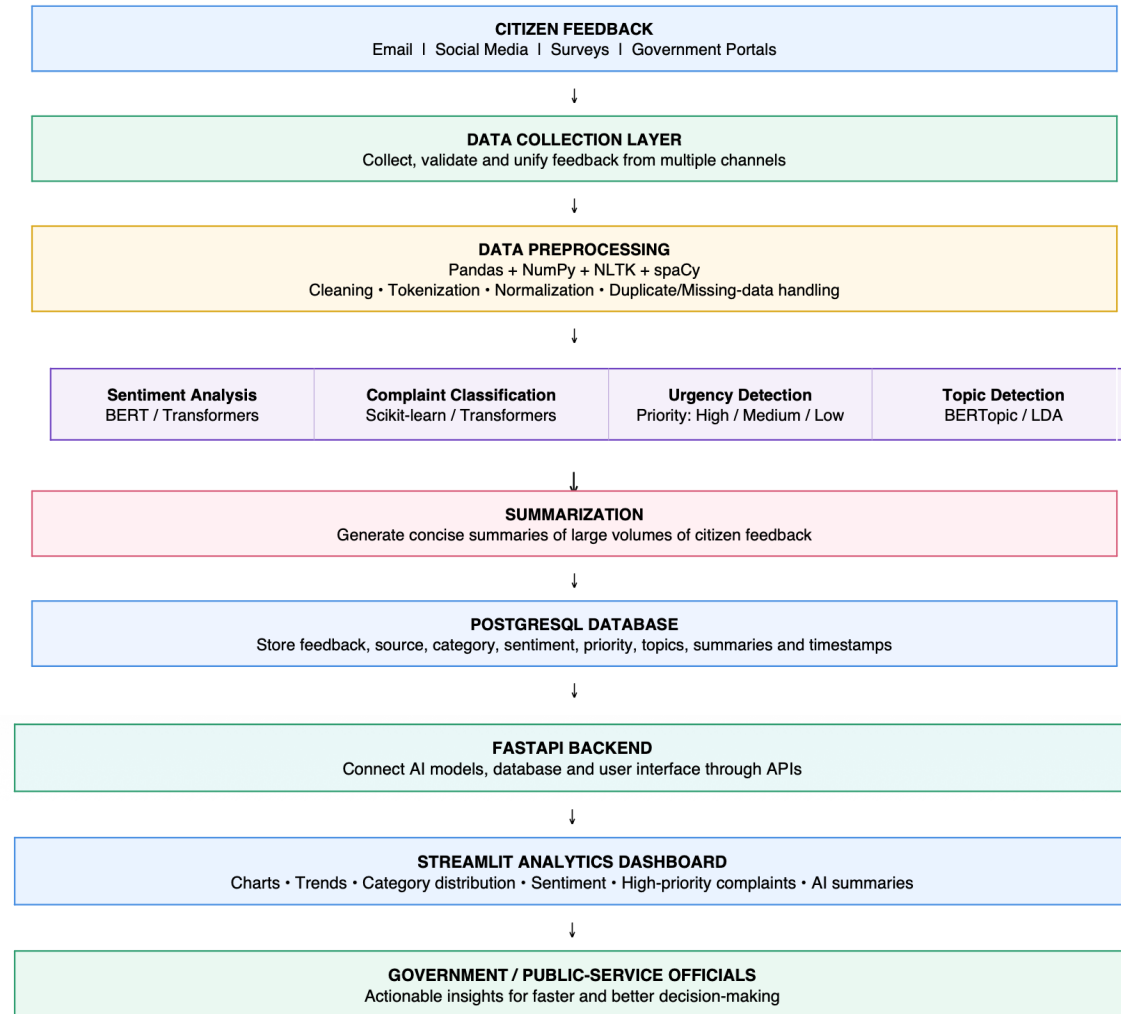
Analysis of Problem Statement (contd...)

Component	Technology	Purpose
Database	MongoDB	Store citizen feedback and analysis results
Data Processing	Pandas, NumPy	Data cleaning and manipulation
Visualization	Plotly, Matplotlib	Graphs, charts, and analytics
Development Environment	Jupyter Notebook, VS Code	Model development and coding
Version Control	Git & GitHub	Source code management and collaboration
Deployment (Optional)	Render / Railway / Streamlit Cloud	Deploy the web application

Analysis of Problem Statement (contd...)

Project Architecture

AI Assistant for Multi-Channel Citizen Feedback Analysis in Public Services

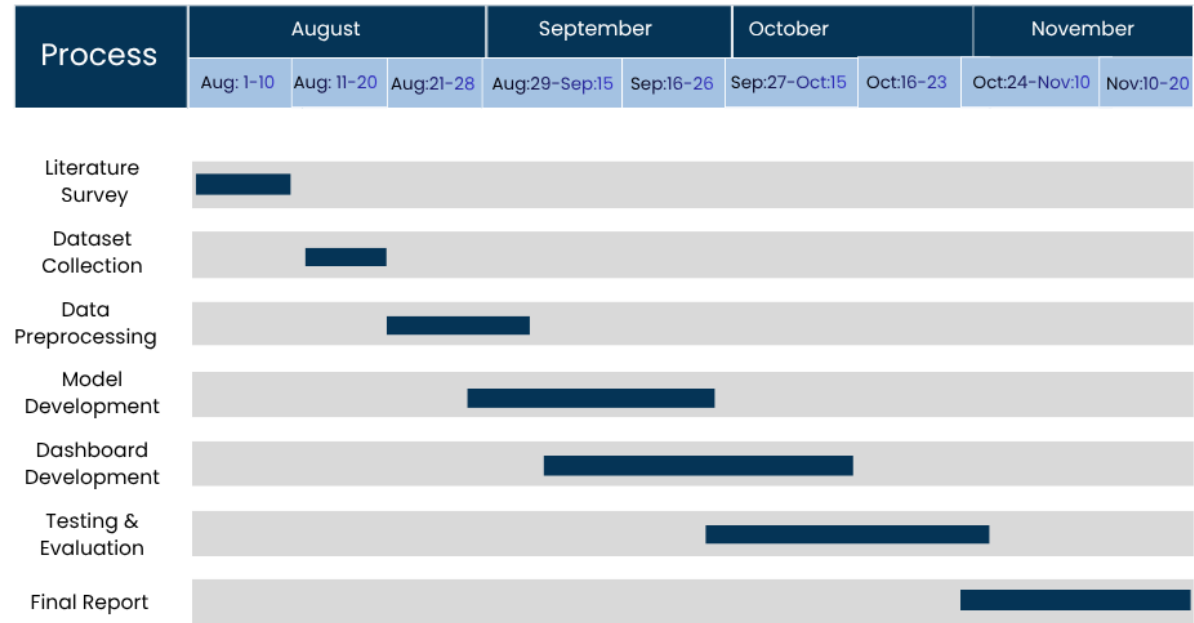


Supporting tools: Python • Git • GitHub • VS Code • Jupyter Notebook • Plotty

Timeline of the Project (Gantt Chart)

PROCESS TIMELINE

Gantt Chart



References (IEEE Paper format)

- Sarvestani, M. (2025). *An NLP-based framework for sentiment and topic analysis of citizen feedback on U.K. government mobile applications*. *IEEE Access*, 13, 210360–210377.
- <https://doi.org/10.1109/ACCESS.2025.3641669>
- Airlangga, M. C., et al. (2023). *Understanding citizen feedback of Jakarta government super app: Leveraging deep learning models*. In *2023 IEEE International Smart Cities Conference (ISC2)* (pp. 1–6). IEEE.
- <https://doi.org/10.1109/ISC257844.2023.10293572>
- Alkraiiji, A. I. (2020). *Citizen satisfaction with mandatory e-government services: A conceptual framework and an empirical validation*. *IEEE Access*, 8, 117253–117265.
- <https://doi.org/10.1109/ACCESS.2020.3004541>



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